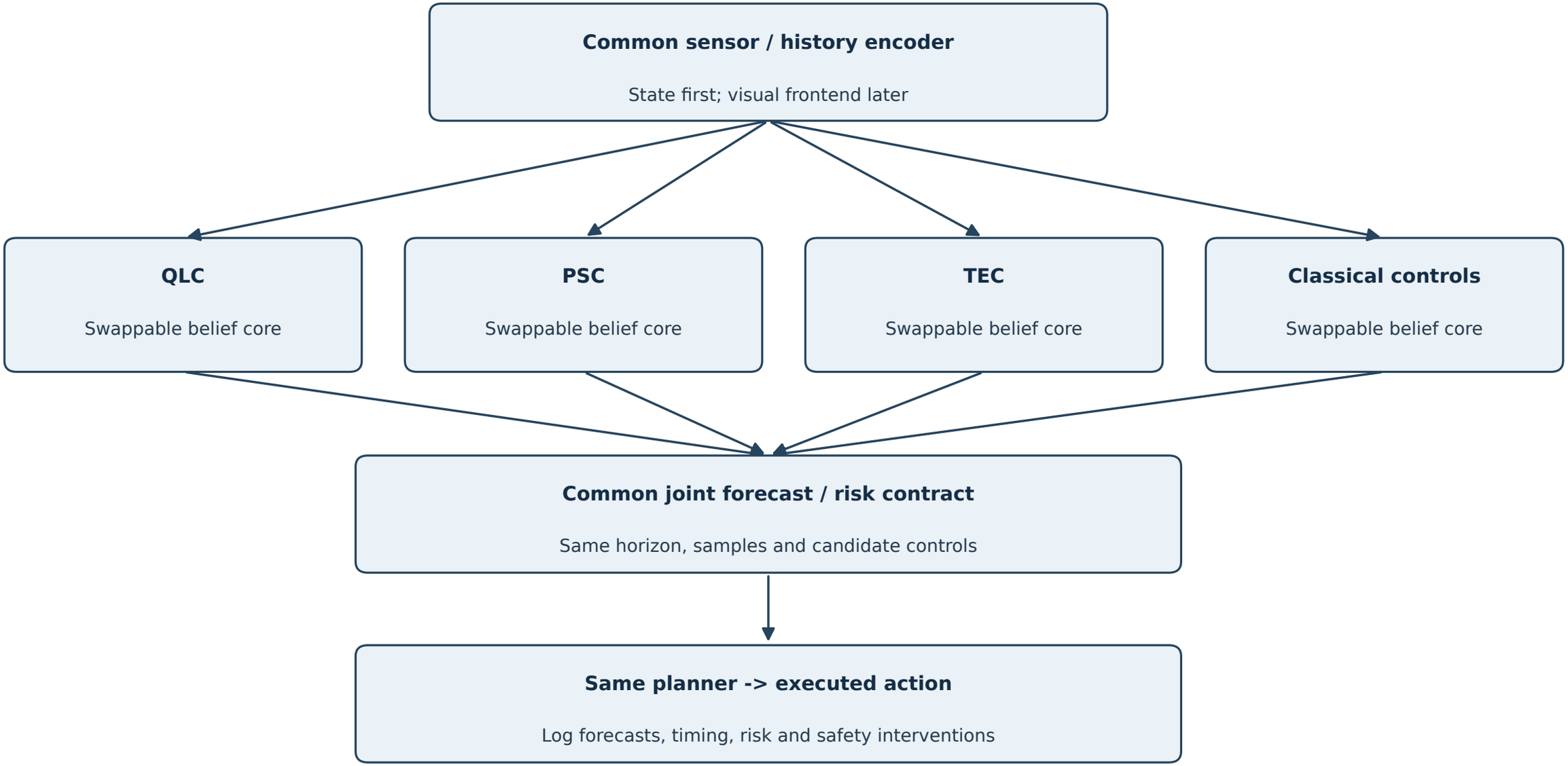
Shared architecture for fair comparison

Keep observation history, decoder contract, objective and control budget comparable.